

EE-608: DIGITAL IMAGE PROCESSING

Lab Assignment 9

Link for sample images : [Sample Images](#)

AIM:

1. a) Write a function to apply a low pass filter to a given image and observe the image and fourier plot before and after.
b) Write a function to apply a high pass filter to a given image and observe the image and its fourier plot before and after.
c) For the two operations above. Verify that convolution in one domain is multiplication in another domain through the program and its plot.
2. a).Write a function for unsharp masking in frequency domain to a given image and observe the image and fourier plot before and after.

b) Write a function for a notch filter(both selection and rejection) to a given image and observe the image and its fourier plot before and after.

c) Write a function for an ideal band pass filter(use frequency mask) to a given image and observe the image and its fourier plot before and after.

Libraries: Opencv to read or write image

Matplotlib to display image

Numpy wherever necessary Basic

python functions

Deliverables:

Just give a concise report of what you understand in your own words, not too heavy theory taken from the web.

- 1) Write a python or matlab code to implement above given tasks.
- 2) Analyze the tasks clearly and the mathematics behind the task.